

1 Linking species traits and vulnerability indicators 2 in European Odonata

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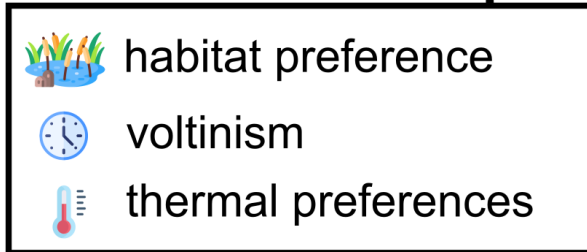
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48 Graphical abstract

vulnerability ~ traits

Red List status
Distribution trend
Area of occupancy



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49 Abstract

50 Understanding the mechanisms and commonalities driving species' vulnerability is
51 essential for prioritizing and guiding conservation efforts. Trait-based approaches
52 offer a mechanistic foundation for generalizing species vulnerabilities within a
53 taxonomic group. Here, we assess how the vulnerability of European Odonata is
54 associated with their traits. Our aim was to (1) quantify the link between traits and
55 vulnerability and (2) identify the most important traits in a multi-trait context. For
56 123 species, we linked 3 vulnerability indicators (Red List categories, distribution
57 trends and areas of occupancy) to a dozen traits, using discriminant and
58 redundancy analyses. We find that 48 to 64% of the variability in vulnerability
59 indicators is explained by traits. The main traits related to vulnerability are habitat,
60 voltinism and thermal preferences. More specifically, vulnerable species tend to
61 associate with oligotrophic habitats or Mediterranean streams. They also tend to
62 have longer life cycles, but this relationship is reversed for species with a small
63 area of occupancy. Species vulnerable because of their decreasing distribution
64 tend to have cold thermal preferences. Vulnerable species generally show a
65 narrow thermal range (except for species vulnerable because of their decreasing
66 distribution). Assessing species' vulnerability is crucial to inform conservation: our
67 trait-based approach provides clues regarding pressures responsible for species
68 vulnerability, thus allowing to plan conservation action targeting groups of species
69 sensitive to the same pressures, rather than focusing on individual species. Our

method provides novel opportunities for predicting species' vulnerability, and
paves the way for building a multi-species conservation indicator for Odonata.

Keywords

Odonata; discriminant analysis; vulnerability; climate change; anthropogenic
pressures; dragonflies

Highlights

- Traits explain up to 64% of the vulnerability status of Odonata
- Odonata vulnerability is linked to habitat, voltinism and thermal preferences
- Preserving or restoring oligotrophic habitats is essential for Odonata conservation
- Mitigating the impact of climate change is crucial for Odonata conservation

Introduction

Anthropogenic activities impact biodiversity in all taxa, but to varying degrees
among and within groups of organisms. These differences in species' response
to environmental changes and human pressures shape local and regional
biodiversity, with certain populations, species or groups of species appearing more
vulnerable than others (Ricciardi and Rasmussen 1999). Determining which

mechanisms are responsible for these differences in vulnerability is paramount to identifying species most at risk and proposing efficient biodiversity conservation measures (Foden et al. 2013; Chichorro et al. 2019).

Species vulnerability is usually framed as a combination of extrinsic exposure to threats, and intrinsic species response (sensitivity and ability to adapt), usually approximated with species traits (Bellard et al. 2025; Foden et al. 2019). For instance, considering the vulnerability of a species to climate change, its extrinsic exposure might be measured with the change in mean temperature over the species' range, and its intrinsic response might be measured with its thermal preference.

As species' response to threats is determined largely by their biological characteristics, we expect links between species' traits and their vulnerability indicators, such as population trends, species range sizes and occupancy rates (Foden et al. 2019; IUCN Standards and Petitions Committee 2024). Traits offer mechanistic insights as to why some species (or populations) are more threatened than others and may in turn help us better anticipate and prevent species declines (Butt and Gallagher 2018). For instance, grassland habitat-specialist butterflies are strongly dependent on grassland habitats quality: as a result, these species are particularly vulnerable to agricultural intensification of grasslands (van Swaay et al. 2020). Using traits and functional metrics also allows comparisons and transferable inference across species or biogeographic areas with different taxonomic composition, but sharing a similar functional composition (McGill et al.

2006), as well as predictions of future responses under scenarios of anthropogenic pressures (Gallagher et al. 2021; Stevenson et al. 2021). For biodiversity conservation, traits may facilitate working with groups of species rather than dealing with species as individual entities.

Many studies have investigated the link between traits and vulnerability, revealing general as well as taxa-specific patterns of associations between traits and vulnerability status (Purvis et al. 2000; Chichorro et al. 2022). These studies have generally found habitat specialization, slow life history traits, small climatic range and low dispersal ability to be linked with higher vulnerability (Chichorro et al. 2022). However, a vast majority of studies have focused on vertebrates. Studies focusing on insects are particularly scarce (only 14% of studies in Chichorro et al. 2019), and even less have investigated this link at continental scales (but see for example Essens et al. 2017 and Bubová et al. 2016, which both focus on European butterflies).

Understanding the drivers of insect vulnerability is extremely important. Indeed, insect populations are facing severe and accelerating declines globally (Cardoso et al. 2024; Lamouille-Hébert et al. 2024). They also represent the vast majority of animal species and are paradoxically understudied (Cooke et al. 2025). Additionally, for aquatic insects, trends can be especially noisy and difficult to assess (Didham et al. 2020; van Klink et al. 2020; Jähnig et al. 2021), with recent data syntheses suggesting complex dynamics of decline followed by an increase that has recently slowed (Environmental Data Compendium 2025; Haase et al.

2023; Outhwaite et al. 2020; van Klink et al. 2020). In this context, species traits offer an alternative approach to guiding knowledge of insect declines.

Here, we contribute to filling the traits-vulnerability knowledge gap in insects by investigating the link between traits and vulnerability in an insect order at the continental scale. We focus on Odonata (dragonflies and damselflies), an emblematic insect order of freshwater habitats. Few studies have investigated the links between traits and vulnerability in Odonata, and they were limited in taxonomic and/or geographic scope (Bowler et al. 2021; Powney et al. 2015; Termaat et al. 2019). Odonata offer a useful model for assessing the relationships between traits and vulnerability indicators (Rocha-Ortega et al. 2023). As adults, they are charismatic and colorful, often easily detected and identified to species level, enabling large accumulations of species-list and point-occurrence data (Bried et al. 2020; Sandall et al. 2022). Moreover, Odonata are often proposed as surrogates for freshwater biodiversity response to environmental change (Kietzka et al. 2023; Pinkert et al. 2023). Finally, Odonata species range from being highly sensitive to broadly tolerant to human pressures, which makes them good candidates to investigate the role of traits in these variations of vulnerability (Samways et al. 2025; Šigutová et al. 2025, 2023).

In this study, we investigate relationships between traits and vulnerability for 123 European Odonata species, testing the potential of traits as a mechanistic basis to help understand and predict species vulnerability. We used three vulnerability indicators (distribution trend, log-area of occupancy and Red List category),

derived from the Red List of European Dragonflies and Damselflies (De Knijf, Billqvist, van Grunsven, et al. 2024). We linked these indicators with twelve species traits derived from the OdonTraits Europe database (De Knijf et al. 2026).

Our study has two aims. First, we want to quantify how well species traits explain vulnerability indicators. We expect species traits to be strongly linked with all vulnerability indicators, due to their prominent role in the species response to anthropogenic pressures (Foden et al. 2019). Second, we want to characterize the links between traits and these indicators. Our aim is to evaluate if previously described relationships (see Table 2 for more details) hold in an integrative analysis where multiple traits are considered, and to rank their relative importance. To our knowledge, such an integrative analysis has never been performed on Odonata. Our results offer the possibility of refining existing conservation status assessments and to group species for more efficient conservation, management, monitoring and forecasting.

Material and methods

All analyses were performed with R 4.5.1 (R Core Team 2024) and are available at <https://doi.org/10.5281/zenodo.17487291> (Nicvert et al. 2026).

2.1 Traits data

We used OdonTraits Europe (De Knijf et al. 2026), a trait database for European Odonata species including all species with populations in Europe (n = 143). After

filtering for species with full coverage of trait data, $n = 123$ species remained. We used twelve morphological, physiological, phenological, behavioral and ecological traits extracted from OdonTraits Europe (see Table 1). As essential biodiversity variables, these traits are particularly relevant to mediate species' response to global changes (Dicks et al. 2024; Kissling et al. 2018; Pereira et al. 2013).

179 *Table 1: Odonata traits overview. This table provides traits categories*
 180 *(morphological, physiological, phenological, behavioral or ecological), their names,*
 181 *a short description and an overview of their range or modalities.*

Category	Trait name	Trait description	Trait ranges / modalities
Morphological	Body size	Obtained from a PCA (see main text)	[-4.78, 5.67]
	Coloration	Main male body coloration	pale, intermediate, dark
Physiological	Thermal preference	Mean of mean annual temperature across species' range	[2.01, 18.28] °C
	Thermal range	Standard deviation of mean annual temperature across species' range	[0.10, 3.92] °C
Phenological	Flight start	Start of the flight period	Early (January-April), medium (May), late (June-July)
	Log flight period length	Log-length of the flight period in months	[0.69, 2.30]
	Voltinism	Duration of the larval life cycle completion	parti- to semivoltine (≥ 2 years), semi- to univoltine (1-2 years), strictly univoltine (1 year), uni- to multivoltine (≤ 1 year)
Behavioral	Flight behavior	Adult activity mode, opposing species that spend most of their active time flying, or staying on a perch	Flier, mixed flier-percher, percher
	Larval behavior	Morphology and hunting behavior of larvae	Burrower, clasper, hider, sprawler
Ecological	Overwintering	Life stage during the winter	Egg, egg-larva (i.e. the first year as egg and the following year(s) as larva), larva, adult
	Oviposition	Oviposition substrate	endophytic (inside plant tissues), exophytic (on plant tissue or on the water surface) or sedimentary (into sediments)
	Habitat	Preferred habitat	eutrophic, oligotrophic, generalist (i.e. wide range of habitats), streams, southern streams (i.e. Mediterranean), temporary waters

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We expect these traits to be linked to species' vulnerability by driving species' response to different kinds of anthropogenic pressures, grouped in the broad categories of land-use change, climate change or pollution (see Table 2 and references therein). For instance, we expect larger species to be more sensitive to rising temperatures because of their low surface to volume ratio making heat exchanges harder (Zeuss et al. 2017).

For body size, the trait database provided measurements of several body parts. To get a single synthetic measure of body size, we performed a principal component analysis (PCA) on four log-transformed pairs of morphological traits (for male and female): abdomen length, body length (abdomen + thorax), exuvia length and hindwing length. We used species coordinates on the first axis of variation (89% of the total variation) as a body size indicator. This synthetic trait is expected to be related to vulnerability due to its influence on heat exchange and dispersal abilities (see Table 2).

Table 2: Main hypotheses on the relationships between traits and pressures (land-use change, climate change or pollution).

Trait	Pressure	Broad mechanism	Hypothesis	Supporting reference(s)
Body size	Climate change	Heat resistance	Bigger species have less efficient heat exchange (Bergmann's rule)	Acquah-Lampitey et al. 2020 Zeuss et al. 2017
	Land use change	Dispersal	Bigger species have stronger flight muscles, which helps them disperse more easily	de Resende et al. 2021
Coloration	Climate change	Heat resistance	Darker species are more sensitive to heat	Acquah-Lampitey et al. 2020 Moore et al. 2019
Thermal preference	Climate change	Heat resistance	Warm-adapted species benefit from global warming, contrary to cold-adapted ones	Bowler et al. 2021 Šigutová et al. 2025
Thermal range	Climate change	Heat resistance	Species with a broader tolerance are less sensitive to variations in temperature accentuated by climate change	Rocha-Ortega et al. 2020
Flight start	Climate change	Resistance to extreme events Response to phenology cues	Species can be affected differently by climate change depending on their phenology timing	Li et al. 2025 Forrest 2016
Flight period length	Climate change	Resistance to extreme events	Longer flight period favors resistance to extreme events	Bubová et al. 2016
Voltinism	Climate change	Adaptability	Species which have more generations per year evolve, and thus adapt, faster	Braune et al. 2008 Viza et al. 2023
	Climate change	Resistance to extremes events	Multivoltine species are also less sensitive to extreme climatic events	Forister et al. 2018
Flight behavior	Climate change, land-use change	Dispersal	Fliers tend to disperse more easily	Angelibert and Giani 2003
	Climate change	Heat resistance	Fliers tend to be endothermic, so they are less sensitive to coldwaves under climate change	Ferreira et al. 2023

Larval behavior	Climate change	Drought resistance	Burrower larvae can survive longer drought periods	Pernecker et al. 2020
Overwintering	Climate change	Phenology	Species overwintering as eggs are less able to advance their phenology	Hassall et al. 2007
	Climate change	Adaptability	Species overwintering as eggs can also adapt to droughts	De Block et al. 2008
Oviposition	Climate change, Pollution	Resistance to pollutants Heat resistance	Oviposition substrate mediates exposure to disturbances: eggs deposited in water are more sensitive to pollution or droughts, while eggs deposited on plants tissue are more sensitive to heatwaves	Angert et al. 2011
Habitat	Land use change, climate change	Dispersal	Lotic species tend to have lower dispersal abilities, which makes them more sensitive to localized habitat destruction or extreme climatic events	Acquah-Lampsey et al. 2020 Rocha-Ortega et al. 2020
	Pollution	Resistance to pollutants	Oligotrophic specialists are more sensitive to eutrophication	Powney et al. 2015 Bowler et al. 2021
	Climate change	Drought resistance	Species adapted to temporary waters are more resistant to droughts	Schiel and Buchwald 2015

2.2 Vulnerability indicators

We related traits to three different indicators of species vulnerability, which were obtained from the European Red List of Odonata (De Knijf, Billqvist, van Grunsven, et al. 2024): distribution trend category, log-area of occupancy and the resulting Red List category (according to IUCN-criteria). These indicators provide complementary insights: distribution trend reflects recent vulnerability (here, over the last 10 years); log-area of occupancy reflects vulnerability resulting from longer time scales processes; and Red List category an integrative measure, which is expected to partly relate to the other two indicators (see tests in in Section 2.3).

The distribution trend category, coded as increasing, stable or decreasing, was derived from the slope of a generalized linear model estimating the rate of change of species occurrence in gridded 10×10 km squares between 2010 and 2020 (for details, see De Knijf, Billqvist, van Grunsven, et al. 2024). Here, we assume that a decreasing trend is an indicator of higher vulnerability.

Species' area of occupancy was estimated as the extent of gridded spatial units (2×2 km squares) where the species was recorded between 2000 and 2020 (see De Knijf, Billqvist, van Grunsven, et al. 2024 for more information). For all analyses, we log-transformed the area of occupancy to make it closer to a Gaussian distribution. Following IUCN standards, we assumed that a smaller (log-)area of occupancy corresponds to increased vulnerability.

Species' European Red List category is an indicator of species' extinction risk at the European scale (IUCN Species Survival Commission 2012). Here, the least to the most vulnerable categories are: Least Concern, Near Threatened, Vulnerable, Endangered and Critically Endangered. For most species, these categories are based on species distribution trend and area of occupancy (computed as explained above), and on population size information when available (see De Knijf, Billqvist, van Grunsven, et al. 2024).

2.3 Relationships between vulnerability indicators

As a preliminary check, we investigated whether the different vulnerability indicators were linked by testing the pairwise links between (1) Red List and log-area of occupancy, (2) Red List and distribution trend and (3) distribution trend and log-area of occupancy (see Appendix A for more details). We expect the Red List status to be linked to the two other indicators, as they are used for assessing Red List status, but there is no a priori reason for trend and log-area of occupancy to covary.

As expected, Red List Status is strongly correlated with log-area of occupancy ($F_{4,118} = 5.914$, $p = 0.0002$; large effect size $\eta^2 = 0.17$, $CI_{95} = [0.06, 1.00]$). Red List status is also strongly linked with distribution trend ($\chi^2 = 101.5$, 2000 permutations, $p = 0.0005$; large effect size $V = 0.62$, $CI_{95} = [0.48, 1.00]$).

Contrary to our expectations, distribution trends are moderately related to log-area of occupancy ($F_{2,120} = 3.511$, $p = 0.033$; medium effect size $\eta^2 = 0.06$, $CI_{95} = [0.00, 1.00]$).

2.4 Linking traits and vulnerability indicators

To assess the relationship between traits and vulnerability indicators, we used multivariate analyses which identify species' functional space explaining the most variation in vulnerability indicators. Contrary to univariate methods (e.g. (generalized) linear models), these multivariate methods are well-suited to identify the most important trait combinations underlying vulnerability, even with multiple correlated traits.

To characterize the link between traits and the two categorical vulnerability indicators (distribution trend and Red List category), we used discriminant analysis (Rao 1948), which maximizes the separation between indicator classes in the multivariate functional space defined by traits (*discrimin* function from the *ade4* R package, Dray and Dufour 2007).

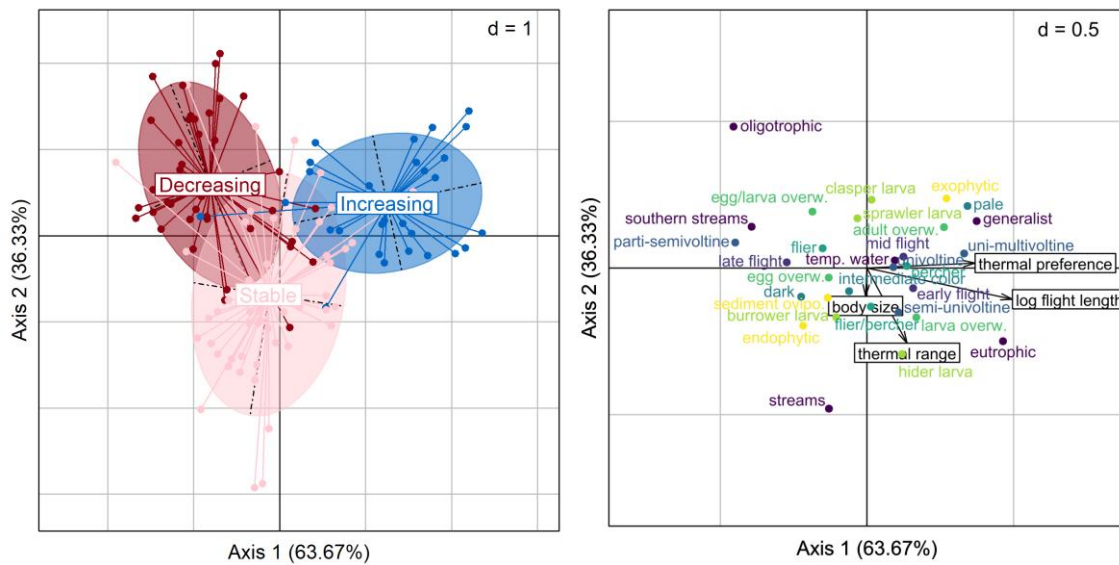
Second, to investigate the link between traits and log-area of occupancy, which is a continuous variable, we used redundancy analysis (Rao 1964; van den Wollenberg 1977). Redundancy analysis finds the functional space in which the variance of the response variable is maximized. Here, since the response is only one variable (log-area of occupancy), the space is only in one dimension, defined by a linear combination of traits.

To quantify the part of variation in vulnerability indicators explained by species traits, we built three (generalized) linear models. These models relate each vulnerability indicator with species coordinates on the functional axes of the multivariate analyses (respectively, one, two or four axes for the log-area of occupancy, the distribution trend and the Red List category). For species' distribution trend and Red List category, we evaluated the part of variation explained by traits using McFadden's pseudo- R^2 of a multinomial regression model (McFadden 1987). For the log-area of occupancy, we used the Pearson R^2 of the linear model. To get confidence intervals on these R^2 , we bootstrapped the data with 999 resamples.

Results

3.1 Distribution trends

The different categories of distribution trends (decreasing: $n = 40$; stable: $n = 52$; increasing: $n = 31$) are well-separated in the multivariate space. They occupy roughly the same proportion of the multivariate space (Figure 1a; see Appendix B: Figure B1 for details on each species' position in the discriminant space).



(a) Distribution trends

(b) Traits in the functional space

275 Figure 1: Discriminant analysis on distribution trends. The percentages associated
 276 to the axes represent the part of variation explained in the constrained discriminant
 277 space. (a) Species grouped according to distribution trends in the discriminant
 278 plane. Each dot represents a species, and they are grouped and colored according
 279 to their distribution trend. (b) Traits in the functional space. Quantitative traits are
 280 represented as labelled black arrows, and modalities of qualitative traits are
 281 represented as labelled dots (modalities of the same traits have the same color).
 282 d represents the size of the grid.

283 Species with increasing trends often prefer eutrophic habitats or are habitat
 284 generalists, and are associated with higher temperatures (Figure 1b). These
 285 species also exhibit pale colors, longer flight periods, shorter life cycles (uni- to
 286 multivoltine) and exophytic oviposition (i.e. oviposition on plant tissue or on the

water surface). In contrast, species with decreasing trends are associated with oligotrophic conditions or southern streams, lower temperatures, exhibit darker colors, longer and later flight periods and longer life cycles (parti- to semi-voltine). Finally, species with stable trends tend to occupy streams, tolerate a wide range of thermal conditions and exhibit the “hider” larval strategy (i.e. covering their flattened bodies with sediment or detritus).

According to the multinomial regression, traits explain 51% of the variation in species distribution trends (McFadden’s pseudo- $R^2 = 0.51$, confidence interval at 95% (CI95) = [0.40, 0.68]).

3.2 Area of occupancy

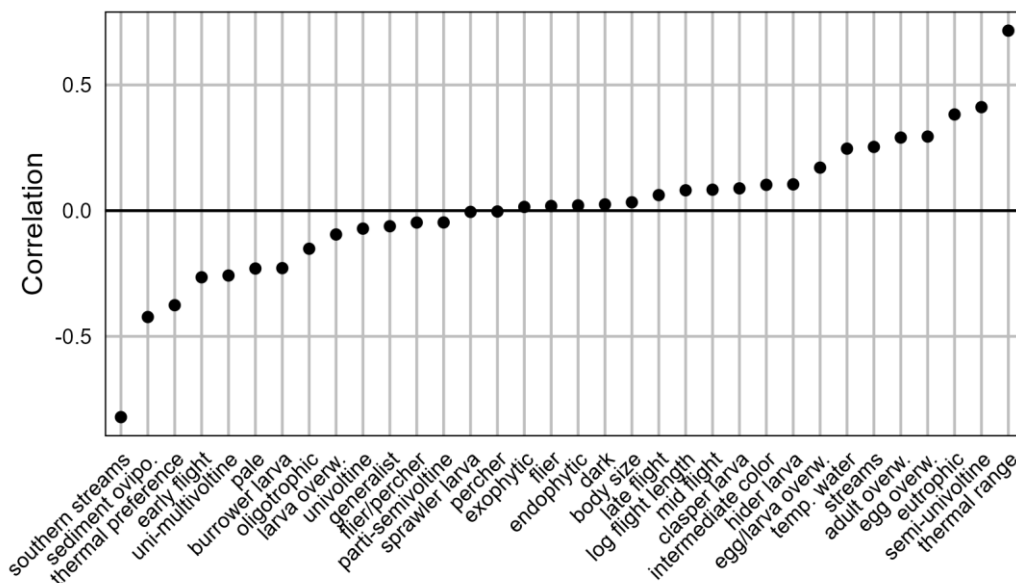


Figure 2: Correlations of traits with the redundancy axis on the log-area of occupancy. The axis is positively correlated with the log-area of occupancy.

Species with small areas of occupancy are primarily linked with southern streams (Figure 2; see also Appendix B: Figure B2). Small areas of occupancy are also weakly associated with species performing sediment oviposition and preferring high temperatures. Larger areas of occupancy are associated with broad temperature tolerance, and to a lesser extent preference for cold temperatures, semi-univoltinism and eutrophic habitat preference (Figure 2).

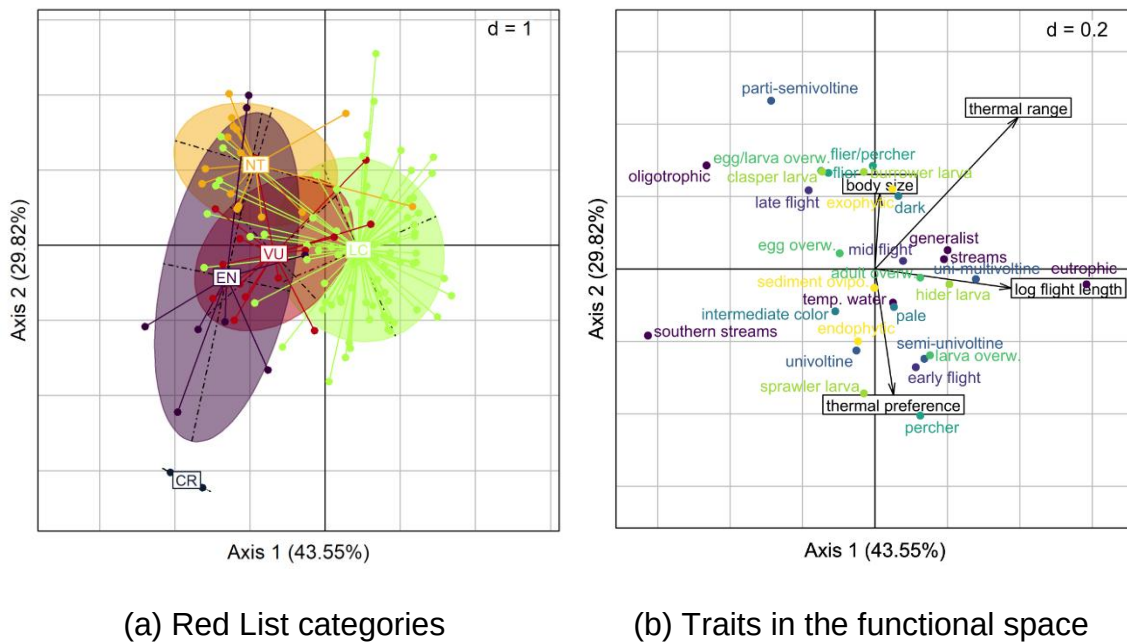
The linear model suggests that traits explain 64% of the variance of the log-area of occupancy ($R^2 = 0.64$, $CI_{95} = [0.53, 0.74]$).

3.3 Red List categories

Red List categories overlaid on the trait space reveal varying degrees of overlap and separation among categories (Figure 3a; see also Appendix B: Figure B3). Endangered species (EN, $n = 8$) are spread out along axis 2, and they overlap extensively with Vulnerable (VU, $n = 14$) and Near Threatened (NT, $n = 15$) species. In contrast, Least Concern species (LC, $n = 84$) occupy a relatively concentrated space for such a large group, and overlap only marginally with the other Red List categories. The two Critically Endangered species (CR) have a unique position in the trait space, close to Endangered (EN) species. Their strong separation from other species is likely due to their very small number. Axis 2 defines a gradient from Near Threatened (NT) to Critically Endangered (CR) species, going through Endangered (EN) species (Figure 3a).

320 Threatened (EN, VU, CR) or Near-Threatened (NT) categories are most linked to
 321 southern stream or oligotrophic habitat preferences, low thermal tolerance and
 322 short flight period (left side of axis 1 in Figure 3b), whereas the Least Concern (LC)
 323 category is most linked to eutrophic habitats, high thermal tolerance, extended
 324 flight periods, and uni- to multivoltine life cycles (right side axis 1 in Figure 3b). The
 325 Critically Endangered and some endangered species also have low values on Axis
 326 2, and they are characterized by percher adults, sprawler larvae, and higher
 327 preferred temperature (Figure 3b).

328 According to the multinomial model, 48% of the variance in Red List categories is
 329 explained by traits (McFadden's pseudo $R^2 = 0.48$, CI95 = [0.40, 0.71]).



330 *Figure 3: Discriminant analysis on Red List categories. The percentages*
 331 *associated to the axes represent the part of variation explained in the constrained*

discriminant space. (a) Species grouped according to Red List categories in the discriminant plane. Each dot represents a species, and they are grouped and colored according to their Red List category (LC = Least Concern, NT = Near Threatened, VU = Vulnerable, EN = Endangered, CR = Critically Endangered). (b) Traits in the functional space. Quantitative traits are represented as labelled black arrows, and modalities of qualitative traits are represented as labelled dots (modalities of the same traits have the same color). d represents the size of the grid.

Discussion

In this article, we related species traits to three proxies of species vulnerability: distribution trend, log-area of occupancy and Red List category. Our data-driven approach allowed us to identify trait syndromes which are most associated with vulnerability proxies. These are (1) slow pace-of-life (low voltinism and flight length); (2) habitat preferences, in particular for southern streams or oligotrophic habitats; (3) cold thermal adaptations (preference for low temperatures, dark color). We also find that about half of the vulnerability is explained by species' traits.

In this study, we do not strictly test the causal impact of each trait on vulnerability, nor any potential non-linear relationship between traits and vulnerability indicators. Moreover, although the trait database we used is not exhaustive, we consider a wide variety of species traits.

4.1 Traits related to several indicators

Our different analyses agree on the important role of the habitat: in particular, a preference for southern streams (all indicators) and oligotrophic habitats (Red List category, distribution trends) is linked to high vulnerability, while a preference for eutrophic habitats (all indicators) and habitat generalism (distribution trends) is linked to low vulnerability. The associations between species' trends and their habitat preferences suggest that land use-associated pressures and pollution could be important drivers of species vulnerability (see Table 2 and references therein). Here, it is likely that nutrient inputs from agriculture and urbanization reduce habitat quality for species preferring oligotrophic conditions. Climate change may also play a role, as southern streams and oligotrophic habitats are more vulnerable to climatic extremes such as prolonged droughts.

Voltinism also emerges as an important trait linked to species vulnerability. Lower number of generations per year (parti- to semivoltinism) is linked to higher vulnerability, whereas a faster life cycle (uni- to multivoltinism) is linked to lower vulnerability. These relationships hold for Red List categories and distribution trends, but not for log-area of occupancy. We can interpret voltinism as a pace-of-life indicator, where faster life-cycle completion allows species to respond more quickly to varying pressures. Species producing one or less generation per year are more vulnerable to extreme events and inclement conditions (e.g. prolonged droughts, floods, heatwaves), where fewer cohorts make it harder to maintain their population size (see Table 2).

Red List category and distribution trend also suggest that species with short flight periods are more vulnerable. Similarly, these species could suffer greater consequences if they are exposed to stressors or harsh conditions (see Table 2). Flight period length also correlates with voltinism, with multivoltine species tending to fly earlier and longer.

Thermal preferences (optimum and breadth) have mixed effects on vulnerability. Indeed, high thermal preference is generally associated with reduced vulnerability (see Table 2), but also with small area of occupancy and Critically Endangered species. Species adapted to high temperatures may benefit from climate change through facilitated range expansion (Engelhardt et al. 2022). However, Mediterranean species, which are warm-adapted, are paradoxically highly vulnerable. This can be explained by many of them being confined to small rivers bordered by mature forests with cool water, even during summers, making them vulnerable to climate warming. Old-growth forests are also becoming exceedingly rare across the Mediterranean region (Barredo et al. 2021). Moreover, a broad thermal tolerance seems to relate with lower vulnerability, suggesting species adapted to a broad thermal range will be less susceptible to the increasingly extreme and variable temperatures induced by climate change (see Table 2).

4.2 Part of explained vulnerability

Traits explain about 50% of the variability in indicators (51% for distribution trends, 64% for area of occupancy and 48% for Red List). This strong link between traits and vulnerability is consistent with our initial expectations.

We hypothesize the remaining variation in vulnerability to be largely driven by differences in exposure. Indeed, vulnerability is defined by the combination of extrinsic exposure to threats and intrinsic species' response, i.e. their sensitivity or capacity to adapt to threats (Foden et al. 2019). Because we used traits as proxies of species response and did not quantify exposure levels, the remaining variability could therefore be explained by heterogeneity in species' exposure over their range.

4.3 Specificities of each indicator

Despite some linkages between the three vulnerability indicators, they provided complementary insights. Species distribution trend can be seen as a snapshot of how well a species is doing over recent time periods (Akçakaya et al. 2021), which in our case was the last decade. Therefore, we can expect this measure of vulnerability to be affected by exposure to recent pressures. Traits which are particularly linked with increasing trends (high thermal preference, habitat generalism and pale coloration) could therefore make species less vulnerable to current pressures, in particular climate change and rapid land-use changes.

By contrast, the (log-)area of occupancy is constrained by historical biogeography, barriers to dispersal, metacommunity processes (such as contemporary dispersal rates, environmental filters and biotic interactions) as well as more recent range constrictions due to anthropic pressures. This suggests that the area of occupancy also holds information on pressures from different scales, especially those acting over longer timescales than the ones affecting species distribution trends. In this context, traits specifically associated to a small range (low thermal tolerance, sediment oviposition and high preferred temperature) may make species more vulnerable to range constriction due to land-use and climate change.

Finally, Red List category is an integrative measure of species vulnerability: indeed, it can be seen as an integrative measure of short- and long-term vulnerability, represented respectively by distribution trends and area of occupancy. This can explain why some traits related to Red List category are also related to distribution trends (preference for oligotrophic and southern streams, voltinism), while others are related to log-area of occupancy (broad thermal tolerance and southern streams).

4.4 Comparison with other studies

Other studies have investigated the links between traits and vulnerability indicators, including Odonata studies from Africa (Deacon et al. 2021), Europe (Powney et al. 2015; Suhonen et al. 2022), and North America (Bried and Rocha-Ortega 2023; Rocha-Ortega et al. 2020). Our findings confirm many relationships

found both across taxa and among dragonflies (see also Table 2 and references therein for studies about dragonflies and other insects).

Our study finds habitat specialist to be more vulnerable, especially those restricted to the rarest habitats (i.e. oligotrophic and southern streams specialists). This is consistent with previous findings from (Chichorro et al. 2022), who described habitat specialization as “the only universal predictor [of extinction risk]” across taxa (see also the “Habitat” row in Table 2 and references therein).

Consistent with vertebrate taxa (Purvis et al. 2000; Chichorro et al. 2022), we also find a strong effect of generation length. Indeed, low voltinism is linked to higher vulnerability using the distribution trend and Red List status proxies (see also Table 2, “Voltinism” row, and references therein).

Finally, although body size is commonly found to affect vulnerability with taxa-dependent directions (Chichorro et al. 2019, 2022), we found no effect for Odonata. Body size is hypothesized play a role in heat resistance, which could affect species’ vulnerability to climate change (see “Body size” row in Table 2 and references therein), but we could not detect this effect here (consistent with Chichorro et al. 2022, and with North American dragonflies in Rocha-Ortega et al. 2020).

Our study provides an assessment of the links between traits and vulnerability for an invertebrate order at a continental level, European Odonata, with a wide range of traits. An expert-based report was published to discuss the results from the

recent European Odonata Red List (De Knijf, Billqvist, Van Grunsven, et al. 2024):
our analysis provides quantitative evidence supporting their conclusions, and
emphasizing the need to pay particular attention to oligotrophic and Mediterranean
stream species.

4.5 Conservation applications

A key opportunity offered by our trait-based approach is to approximate the
vulnerability status of a species using their traits, which could be much needed in
cases where the species' population sizes and dynamics are poorly known. In fact,
knowledge of Odonata population sizes and trends is very geographically uneven
(Bried et al. 2020), with 29% of Odonata species listed as Data Deficient worldwide
(Sayer et al. 2025). Our descriptive approach can use traits as an early-warning
signal regarding the status of some species, akin to using geographic range size
as a proxy of status (Bried and Rocha-Ortega 2023). In our case, for example, the
dragonfly *Aeshna crenata* and damselfly *Coenagrion johannsonni* were classified
as having a stable distribution trend yet were positioned closer to decreasing
species in the trait space (see Figure B1 in Appendix B). These species only occur
in Scandinavia and some Baltic states, where sampling intensity is very low and
the effects of climate change are not as severe compared to further south, so their
decline might not have been detected yet. Traits could also be used in predictive
statistical models to infer species' vulnerability status from their traits (as the
machine learning approach proposed by Bland et al. 2015).

Conservation action can also be guided by the evidence on trait importance, since it suggests possible mechanisms that drive species vulnerability. Our study shows that species preferring oligotrophic or Mediterranean streams habitats are particularly threatened: therefore, restoring their habitat (e.g. by decreasing nutrients input and restoring hydrology) seems particularly important. Several traits suggest the influence of climate change as well, which underlines the need for regional and global climate actions for biodiversity conservation, including the application of novel resiliency frameworks (Anderson et al. 2023) and climate adaptive management (Aldous et al. 2011; Kingsford et al. 2011).

Finally, an indicator based on Odonata species sharing similar traits could provide a monitoring tool to track the condition of specific freshwater habitats to evaluate and guide conservation actions. Multi-species indicators have been used for a variety of taxa, especially for birds (Fraixedas et al. 2020). In multi-species indicators, the selected species usually share a trait, most typically habitat specialization (van Swaay and van Strien 2005; Gregory et al. 2019), which reflects an expected common response to a specific anthropogenic pressure. Our study allows to identify which species are vulnerable using different indicators (see Appendix B), and shows that their traits are linked with vulnerability. This provides evidence that traits can be used to support the selection of Odonata species for multi-species indicators, which could track the state of specific freshwater habitats (e.g., oligotrophic habitats).

4.6 Conclusion

In this article, we related species traits of Odonata to three vulnerability indicators: distribution trend, log-area of occupancy and Red List category. We found that around half of the variation in vulnerability indicators among species could be explained by traits. We also identified habitat preference, voltinism and thermal preferences as the most important traits. Our study suggests that traits could be used to approximate or predict the vulnerability of Data Deficient Odonata species, which is especially relevant given that trait information is typically easier to collect than vulnerability indicators based on population trends (Bried and Rocha-Ortega 2023; Cardoso et al. 2011). We also find preserving and restoring oligotrophic habitats and Mediterranean streams, as well as mitigating the impacts of climate change, as key actionable management actions. Finally, our study opens perspectives for a trait-based Odonata indicator to track and evaluate conservation actions.

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Supplementary material

4.7 Relationships between vulnerability indicators

In this appendix, we measure how the three different vulnerability indicators used in this project are correlated across species.

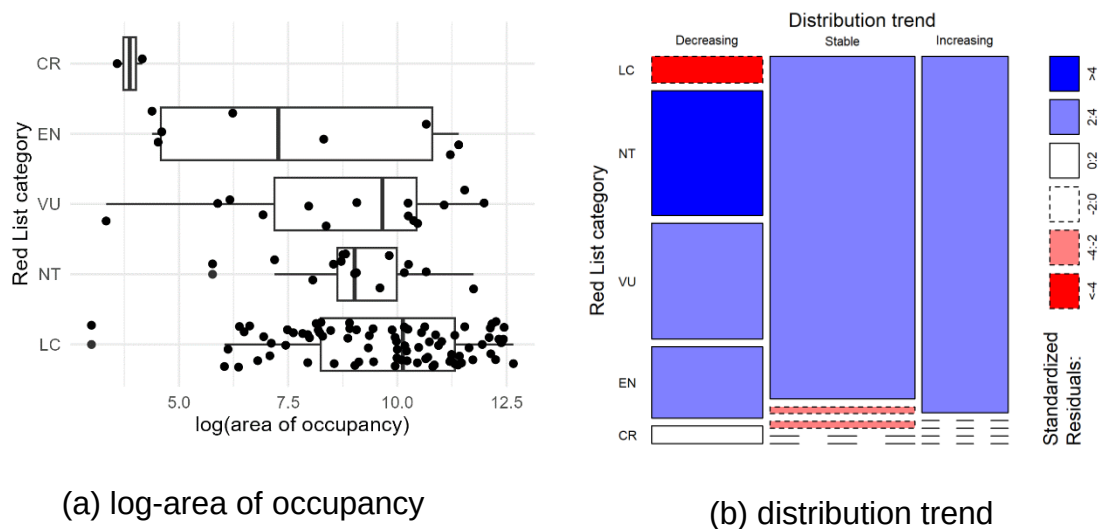


Figure A1: relationship between Red List category and (a) log-area of occupancy and (b) distribution trend. Each square corresponds to the corresponding category on rows and columns, with a size proportional to the observed count. Colors represent the deviation from the expected counts (negative values: observed is lower than expected, positive values: observed is higher than expected).

First, we test the link between Red List category and log-area of occupancy (Figure A1a), using a one-way ANOVA: the link is significant at risk $\alpha = 0.05$ ($F_{4,118} = 5.914$, $p = 0.0002$), with a large effect size ($\eta^2 = 0.17$, $CI_{95} = [0.06, 1.00]$ (Ellis 2010)).

Then, we evaluate the link between Red List categories and distribution trend (Figure A1b). The chi-squared test is significant at risk $\alpha = 0.05$ ($\chi^2 = 101.5$, 2000 permutations, $p = 0.0005$), with a large effect size (Cramér's $V = 0.62$, CI95 = [0.48, 1.00] (Ellis 2010)).

Second, we test the link between log-area of occupancy and distribution trend (Figure A2). We test the link between these variables using a one-way ANOVA: the link is significant at risk $\alpha = 0.05$ ($F_{2,120} = 3.511$, $p = 0.033$), but it is of medium magnitude ($\eta^2 = 0.06$, CI95 = [0.00, 1.00] (Ellis 2010)).

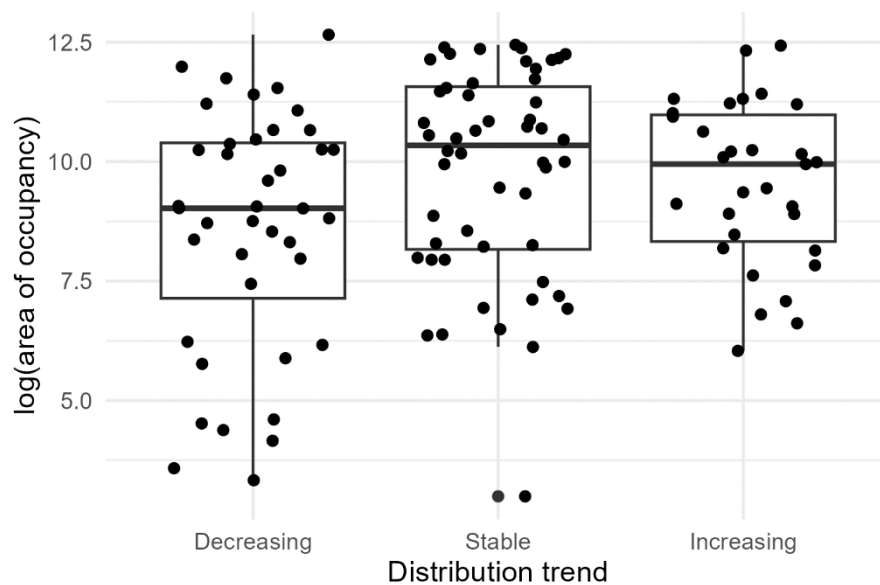
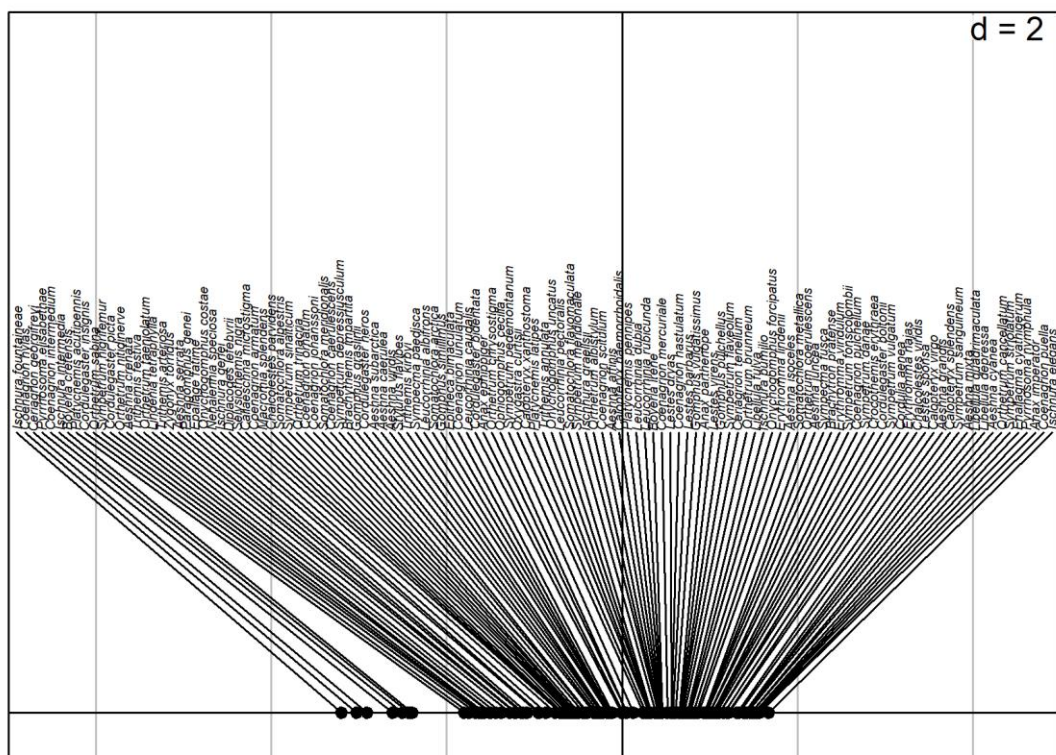


Figure A2: Relationship between distribution trend and log-area of occupancy.



852

853 *Figure B2: species positions on the redundancy axis for log-area of occupancy.*

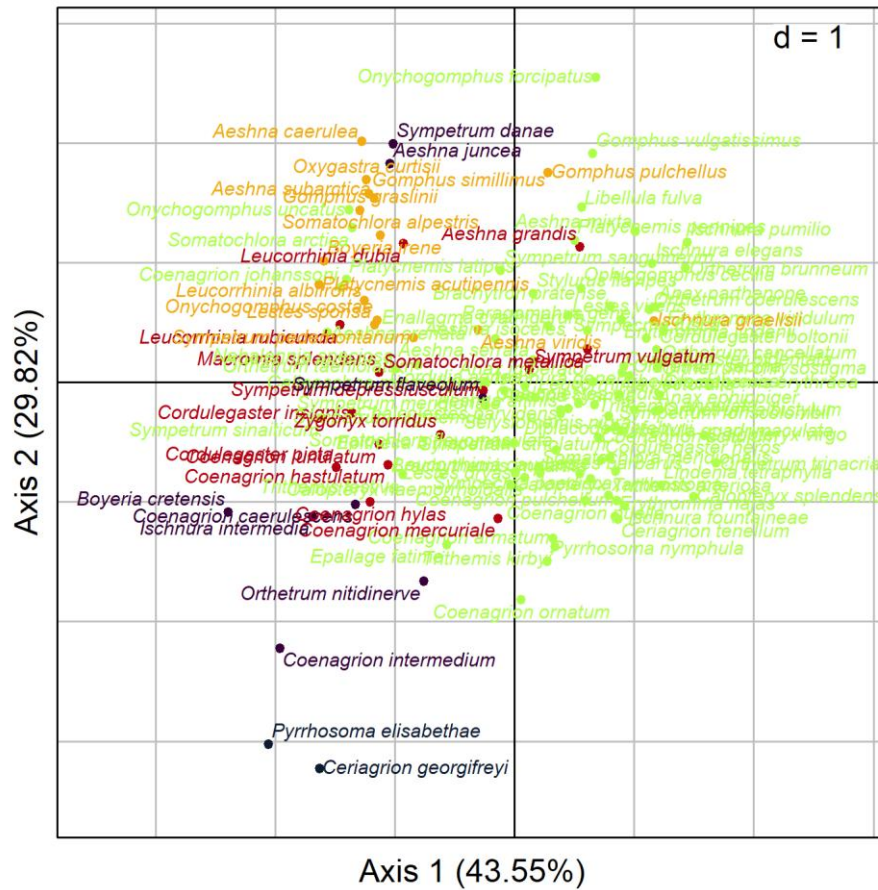


Figure B3: species positions in the discriminant space for Red List status.

857 Supplementary references

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